

Wayfair Rugs Market Intelligence: AI Agent Demo

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Executive Summary

Overview of Market Intelligence Agents

Objective:

Help Wayfair's Rugs Category team automate market intelligence by discovering trends, benchmarking competitors, and generating on-brand marketing content.

My Solution

Built 3 AI agents powering a unified Market Intelligence Dashboard.

- Trend Discovery and Competitor Monitoring analyze market, competitor, and customer insights from Amazon, Walmart, and social platforms.
- AI Insights & Content generates marketing-ready content including product descriptions, blogs, emails, and social media posts.

Core Outputs

- ✓ Trend Discovery Agent - [Sample Output Link](#)
- ✓ Competitor Monitoring Agent - [Sample Output Link](#)
- ✓ AI Insights & Content Agent - [Sample Output Link](#)
- ✓ Integrated Dashboard - [Sample Output Link](#)

Key Insights

Insight 1: Market Opportunity

- Natural fiber rugs growing 28% YoY.
- 47% of searches focus on sustainability.
- Popular materials: Polypropylene, Jute, Recycled PET.

Insight 2: Competitive Landscape

- Ideal price range: \$100–\$200.
- Amazon & Walmart compete on low prices and large inventories.
- Limited focus on customer education and design guidance.

Insight 3: Wayfair Strategy

- Position Wayfair as "the friend who knows design."
- Differentiate through design expertise, sustainability, and educational content.

Agent 1: Moodboard Generator

From Design Prompt → AI-Curated Visual Mood

Objective:

To transform simple text prompts into AI-generated moodboards that reflect emerging rug design styles.

Input Prompt Example

"Bohemian rugs, neutral tones"

Keep in Mind / Input Notes

- Keep prompts **short (under 10 words)** and **specific** — e.g., "eco-friendly jute rugs," "modern geometric rug."
- Use **clear style + material keywords** (e.g., color, texture, tone).
- Avoid subjective adjectives like "beautiful" or "nice."
-  **Free-Tier Limitation:** Google Gemini's image generation API currently allows **up to 20 image generations per day (free version)**.

Output Example:



 [View JSON Workflow](#)

 [View Sample Moodboard](#)

Agent 2: Trend Discovery Agent

Subtitle: From Rug Category → AI-Generated Market Trend Report

Objective:

Analyze market data from Amazon, Instagram, Pinterest, and design blogs to identify emerging rug trends and opportunities to support faster product decisions

Input Prompt Example

Enter a specific rug category or focus area

Sample Input Prompt: **Outdoor Rugs** or

Outdoor Rugs focus: Sustainable

Keep in Mind / Input Notes

- Input Parser1" only matches four hardcoded category phrases (area rug, outdoor rug, hallway runner, shag rug)
- There are many wait nodes added on purpose to slow the workflow down so it doesn't send too many request as we are using free versions of Open Router and Mistral.



Improvements / Expansion

Opportunities

- Add Reddit, Google Trends, Wayfair search/browse data for real buying-intent signals
- Add a vision-capable LLM call (GPT-4V, Claude, or a Mistral vision model) that takes image URLs from Pinterest/Instagram results and extracts style/color/pattern descriptions
- Auto-run weekly with Slack/Email alerts only when a significant trend shift is detected



[View JSON Workflow](#)



[View Sample Output](#)

Agent 3: Competitor Monitoring Agent

Subtitle: From Rug Category → AI-Generated Competitor Intelligence Report

Objective:

Collect and analyze competitor product listings from Amazon and Walmart, benchmark them against Wayfair, and identify pricing strategies, customer sentiment, and market opportunities.

Input Prompt Example

Enter a specific rug category or focus word

Sample Input Prompt: **Outdoor Rugs** or **Outdoor Rugs focus: Sustainable**

Keep in Mind / Input Notes

- Use exact category phrases ("outdoor rug," "area rug," "hallway runner," "shag rug") — otherwise it fails with no_category
- Treat % as directional, not definitive — only 10 products/retailer per run
- Preview full HTML before downloading — tables/cards can render malformed
- Allow 3–5 minutes per run — built-in pacing avoids rate limits
- Check Scope section's product counts first — confirms all 3 retailers returned data



Improvements / Expansion Opportunities

- Could expand to Home Depot, Target, Overstock, or Etsy for a fuller competitive landscape.
- Pre-summarise the full product catalogue (price ranges, materials, counts) instead of feeding the LLM only the first 15 products.
- Review sentiment analysis: Use Apify to scrape reviews → LLM summarizes sentiment/pros-cons per category → deeper insight beyond price/rating.



[View JSON Workflow](#)



[View Sample Output](#)

Agent 4: AI Insights & Content Agent

Subtitle: From Market Intelligence → AI-Generated Marketing Content

Objective:

To transform trend and competitor insights into on-brand marketing content, enabling faster campaign creation and consistent messaging across multiple marketing channels.

Input Prompt Example

Sample Input Prompt:

- Rug Category: Outdoor Rugs
- Focus Area: Natural and Sustainable
- Trend Report: Upload Trend Discovery Report (HTML)
- Competitor Report: Upload Competitor Monitoring Report (HTML)
- Target Audience: Budget-Conscious
- Seasonal Context: Peak Season Launch

Keep in Mind / Input Notes

- Keep inputs specific: Vague "Focus Area"/category entries make the AI improvise more, raising the risk of an incomplete response.
- Don't rename form fields: File matching relies on exact field names. Renaming them means the AI silently gets no report data.
- Check output for completeness (esp. later sections) before sharing, and keep uploaded P2/P3 reports concise since only the first ~15,000 characters are used.



Improvements / Expansion Opportunities

- Split into focused agents (like Trend/Competitor agents) cuts truncation risk, improves quality per section.
- Generate 2-3 variants for subject lines/hooks/headlines (not just one) — lets marketing team A/B test instead of using a single untested version.



[View JSON Workflow](#)



[View Sample Output](#)

Market Intelligence Dashboard

Bringing All Agents Together → Live Insights for Wayfairs' Rugs Team

Objective:

To consolidate Trend Discovery and Competitor Monitoring reports into a single HTML dashboard, providing a centralized view of market intelligence for faster business decisions.

How This Works

- Takes P2 (Trend) + P3 (Competitor) HTML reports as input
- Fetches a blank dashboard template from the backend
- Parses both reports with custom code (no AI) to extract key data points
- Merges extracted data into the template
- Outputs one combined, downloadable dashboard

Keep in Mind / Implementation Notes

- Keep category name consistent with P2/P3 — mismatches won't break parsing, but will make the header inconsistent with the data shown.
- If a section looks empty, check the source report (P2/P3) first — this workflow only extracts what's already there, so missing data usually means it wasn't in the original report, not a bug here.



Improvements / Expansion Opportunities

- Auto-trigger from P2/P3 completion and auto-extract the category name — removes manual upload and mismatch risk.
- Instead of showing empty sections silently, add a validation step that flags what failed to extract so the user knows to check the source report.
- Add swappable dashboard templates and make it interactive using filters, expandable sections, and clickable charts.



[View JSON Workflow](#)



[View Dashboard](#)

Reflections & Future Improvements

What I Learned and How I Can Make It Better



Key Learnings

- I learned that splitting one big AI task into smaller, specialized agents gives more reliable results than one prompt doing everything."
- "I learned output quality depends heavily on which model runs each step — some steps quietly used weaker models than others."
- "I learned most of my time went into getting data to flow correctly between nodes, not into the AI prompts themselves



Future Improvements / Next Steps

- "I want to connect my agents end-to-end, removing the manual re-upload step between workflows."
- "I want to diversify data sources beyond Pinterest/Instagram/Amazon — adding Google Trends and Wayfair's own search data."
- "I want to add consistent error handling across all four workflows, so failures get flagged, not silently incomplete."

Thank you